

A Machine Learning Perspective for the Optimization of Annealing Parameters in Solution Processed Thin Film Devices

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Abstract

This work adopts machine learning to refine the annealing process in solution-processed thin films, such as QDs, perovskites, and organic semiconductors. The annealing temperature and time of PbS QD thin film is optimized using a machine learning algorithm. This method offers a systematic approach to determine the optimal annealing parameters, surpassing the traditional, less-efficient methods. This adaptive model minimizes human bias, handles complex nonlinear relationships, and capable of optimizing multiple parameters simultaneously. By introducing machine learning algorithm, this method provides a universal and effective strategy for the optimization of parameters during the fabrication of thin films.

Author Keywords

Machine learning; quantum dot; annealing parameters.

1. Introduction

In the field of optoelectronics, solution-processed thin film devices, including quantum dots (QDs), perovskites, and organic semiconductors, have gained prominence due to their unique optoelectronic properties. Quantum dots, for instance, exhibit size-dependent bandgaps, while perovskites are known for their high absorption coefficients [1,2]. These materials are fabricated using various techniques such as spin-coating and inkjet printing, each playing a crucial role in defining the film's morphological and interfacial characteristics [3].

The annealing process is a pivotal step in these fabrication techniques, enhancing the film's microstructure and electronic properties, thereby improving device performance [1,3]. Traditional annealing methods, however, often lack efficiency, leading to suboptimal parameter discovery due to broad temperature gradients and sparse sampling [4]. This inefficiency calls for a novel approach to parameter optimization.

Machine learning emerges as a transformative solution, offering adaptability and precision in optimizing annealing parameters [5,6]. This study harnesses machine learning algorithms to refine parameters continuously with new data, reducing human bias and ensuring consistent results. These algorithms excel in modeling complex, nonlinear relationships, a task challenging for traditional methods [7,8].

Furthermore, machine learning models can optimize multiple parameters simultaneously, considering their interdependencies, a significant advantage over conventional methods [5,6]. This data-driven decision support aids in identifying optimal process parameters, enhancing product quality and production efficiency. The scalability and versatility of this method make it applicable across various industrial settings.

2. Result and Discussion

Experimental Idea and Algorithm Introduction

Our investigation into PbS quantum dot photodetectors demonstrates that annealing significantly improves photocurrent responses under infrared illumination. The machine learning model optimized annealing parameters within a 50°C to 130°C temperature range and 0 to 40 minutes duration, outperforming traditional methods [2,3]. Specifically, conditions such as 106°C for 31.2 minutes were determined to improve the responsivity, not only exceeding the baseline of 500 mA/W, but also improving the responsivity before and after annealing by more than 230% [1].

The integration of Ant Colony Algorithm (ACO) and hill-climbing algorithm (HC) further refined parameter identification, ensuring greater accuracy and efficiency in photodetector production [5,6]. This methodology not only promises enhanced accuracy and efficiency in optoelectronic device production but also suggests broader industrial applicability and long-term cost benefits [7,8]. The machine learning approach, with its real-time optimization and feedback capabilities, is well-suited for fast-paced production environments, offering immediate optimization suggestions in response to new data [4].

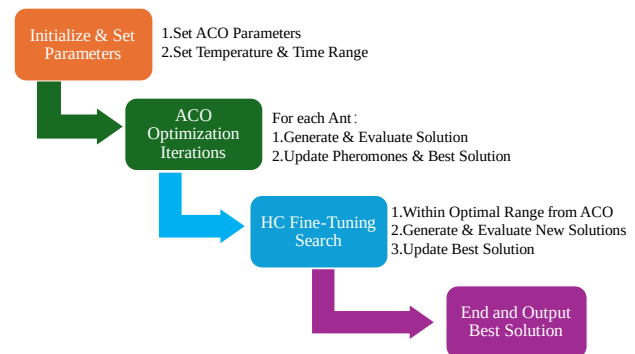


Figure 1. Ant Colony Algorithm and Hill-Climbing Optimization Process.

The provided flowchart outlines the steps of a ACO process enhanced with a HC for refinement, as shown in Figure 1. It begins with the initialization of the swarm, setting parameters for temperature and time, and then moves through cycles of measuring responsivity, updating bests, and adjusting velocities and positions. A decision point checks if optimal conditions have been met, at which point hill-climbing is applied for further refinement if necessary. The process concludes once the optimal temperature and time parameters are identified, ending the whole process.

Pseudocode for ACO-HC Hybrid Optimization

Algorithm 1 Core Logic of ACO and HC Optimization

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1: for each ACO iteration do
2:   for each ant do
3:     Generate solution
4:     if solution is better then
5:       Update best solution
6:     end if
7:   end for
8: end for
9: Use best ACO solutions to define HC search range
10: for each HC iteration do
11:   Generate new solution within ACO range
12:   if new solution is better then
13:     Update best solution
14:   end if
15: end for
16: Output: Best solution from HC

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The integration of ACO and HC within the proposed optimization algorithm presents significant advantages for tackling complex optimization challenges. This synergy enhances exploration capabilities, as ACO's collective intelligence navigates multiple solution paths, reducing the likelihood of settling on suboptimal solutions prematurely. Following this, HC refines these solutions by conducting a focused search within promising areas highlighted by ACO, ensuring the solution space is thoroughly exploited for the best possible outcomes. This hybrid approach capitalizes on ACO's broad search capabilities and HC's local optimization precision, achieving superior solutions unattainable by either algorithm alone. Furthermore, the algorithm's adaptable nature permits parameter and strategy adjustments to suit specific problem nuances, broadening its utility across various optimization tasks. Its robust performance, consistent across diverse problem landscapes, underscores the algorithm's resilience to the inherent variability of optimization problems.

Device Fabrication Steps

The devices used for our tests are PbS quantum dot-based conductor devices fabricated by the LBL method. Details of the device fabrication method are shown in Figure 2.

The step of LBL (layer by layer) is to prepare a layer of PbS quantum dot film on a piece of clean ITO, drop a small amount

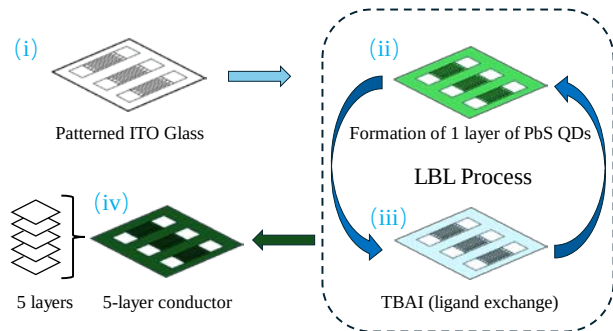


Figure 2. The fabrication process flow of photodetectors.

of 10mg/ml TBAI solution for ligand exchange, and then proceed to the next layer of film preparation after the ligand exchange is completed.

By manipulating the LBL in a way that ensures consistent PbS quantum dot material, we can obtain benchmark devices with roughly the same thickness and small performance errors.

Test Methods and Test Conditions

The complete test flow of this work is shown in Figure 3, firstly the fabricated conductor device is tested for the magnitude of light-dark current at -20 V to 20 V using APDQE at 530 nm with incident optical power P of 40 μ W and the base responsivity is calculated. After that, annealing experiments with different parameters were carried out, the annealing atmosphere was air and the annealing equipment used was BK946 manufactured by BAKON. After annealing wait for natural cooling to room temperature. The light and dark current was tested again under the same conditions and the responsivity after annealing was recorded. The change in responsivity before and after was used to calculate the responsivity enhancement rate at that annealing temperature and annealing time, as is shown in Figure 3.

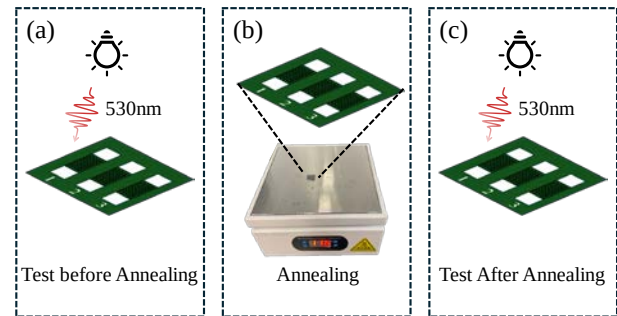


Figure 3. The flow of the test.

The equation for calculating the degree of influence based on the magnitude of the light-dark current is shown in Equation 1

$$R = \frac{I_L - I_D}{P_a} \quad (1)$$

Where, R is the responsivity of the conductor at a single voltage, I_L is the photocurrent, I_D is the dark current and P_a is the actual incident optical power.

The equation for calculating the responsiveness improvement rate is shown below:

$$r = \frac{R_A - R_B}{R_B} \quad (2)$$

Where, r is the enhancement rate of responsivity before and after annealing, R_A is the responsivity after annealing and R_B is the responsivity before annealing.

All the devices followed this flow of the test and the experimental data can be considered real and reliable.

Moreover, this procedure allows us to accurately measure the response enhancement of the conductor devices before and after annealing. The APDQE instrument and the equipment used for annealing are shown in Figure 4.

Data collection

Based on the detailed preparation method in Figure 2 and the test procedure in Figure 3, we recorded the data shown in Table 1:

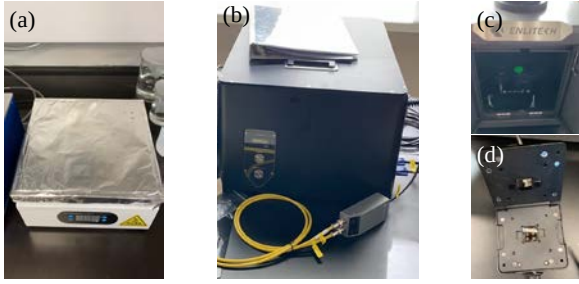


Figure 4. The annealing equipment(a), APDQE's light source(b), APDQE's dark box(c), and APDQE's test box(d).

In Table 1, each bracket symbolizes a data point, with the left value indicating temperature and the right value indicating time, for a total of 24 data points.

Table 1. Experimental recorded data

Time(min) \ Temperature(°C)	10	20	30	40
50	(50,10)	(50,20)	(50,30)	(50,40)
60	(60,10)	(60,20)	(60,30)	(60,40)
85	(85,10)	(85,20)	(85,30)	(85,40)
95	(95,10)	(95,20)	(95,30)	(95,40)
110	(110,10)	(110,20)	(110,30)	(110,40)
125	(120,10)	(125,20)	(125,30)	(125,40)

Data Processing and Presentation of Results:

As shown in Figure 5, an ITO has three complete conductor structures on it, i.e., three devices, so we calculate the average value based on the degree of responsivity enhancement before and after the annealing of the three devices as the corresponding enhancement rate after this annealing temperature and time treatment to exclude interference as much as possible.

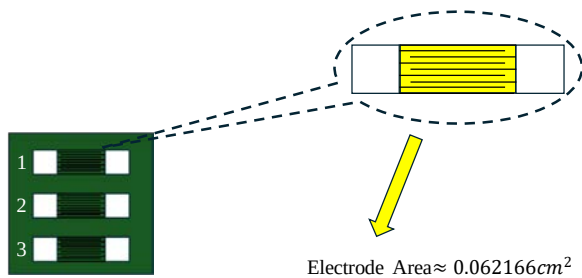


Figure 5. The Schematic structure and Electrode area of the device.

Since the light out of the APDQE is uniform, the number of photons received per unit area is the same. Then there is the following equation:

$$P = P_a \quad (3)$$

From the responsivity enhancement rates before and after annealing of the three devices, the average enhancement rate is

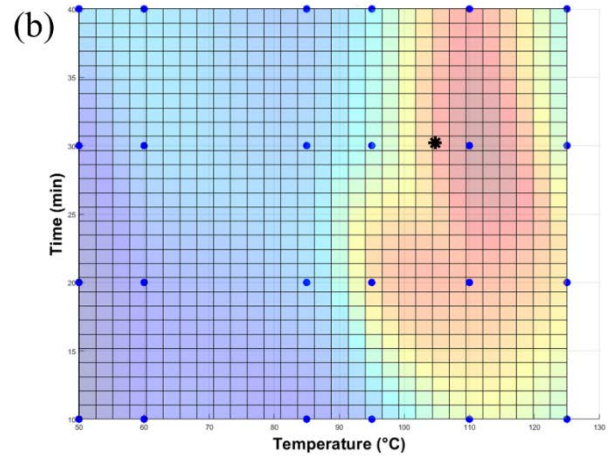
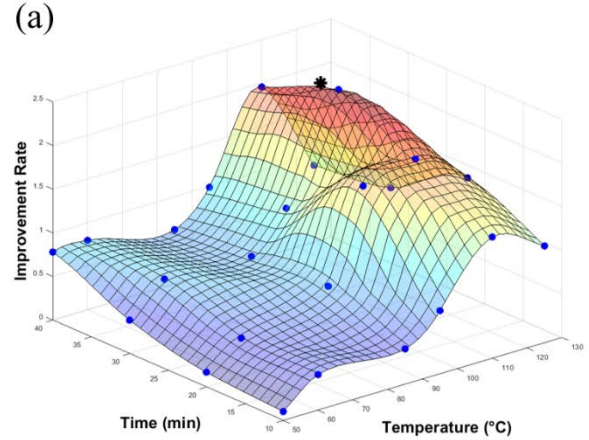


Figure 6. A three-dimensional presentation of the predicted results (a), and a two-dimensional presentation of the predicted results (b).

obtained from the following equation:

$$\bar{r} = \frac{r_1 + r_2 + r_3}{3} \quad (4)$$

Where, $\bar{r}$ is the average enhancement rate, r_1 is the enhancement rate obtained for the first channel device, r_2 is the enhancement rate obtained for the second channel device and r_3 is the enhancement rate obtained for the third channel device.

Utilizing Equations 1-4 and the gathered data, we calculated the responsivity enhancement rates across various temperatures and times, which were then integrated into the algorithm for prediction, as depicted in Figure 6. The data were visualized in 3D using MATLAB, with temperature on the X-axis, time on the Y-axis, and the responsivity enhancement rate on the Z-axis.

As shown in Figure 6, the black asterisk symbols are the optimal parameter points calculated by the code. The blue blob represents the collected data.

In Figure 6, the closer the color of the curve is red, it means that annealing at the temperature and time corresponding to that point can lead to greater responsiveness improvement, and the opposite is true the closer the color is to blue.

Our analysis demonstrates that utilizing the new algorithm offers several advantages over traditional methods. Firstly, it enables the

finding of optimal parameters in finer intervals with a high degree of freedom. Secondly, by combining the ant colony algorithm with the hill climbing algorithm, it overcomes the common drawback of traditional methods, which often get trapped in local optima. This integration not only speeds up the process of finding optimal parameters but also makes it more efficient. Lastly, the new algorithm boasts superior generalizability, making it adaptable for the discovery of higher-dimensional parameters, thereby broadening its applicability and effectiveness in various contexts.

3. Acknowledgement

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